

# Logic or Logics?

**Andrés Sicard-Ramírez**

Formal methods seminar

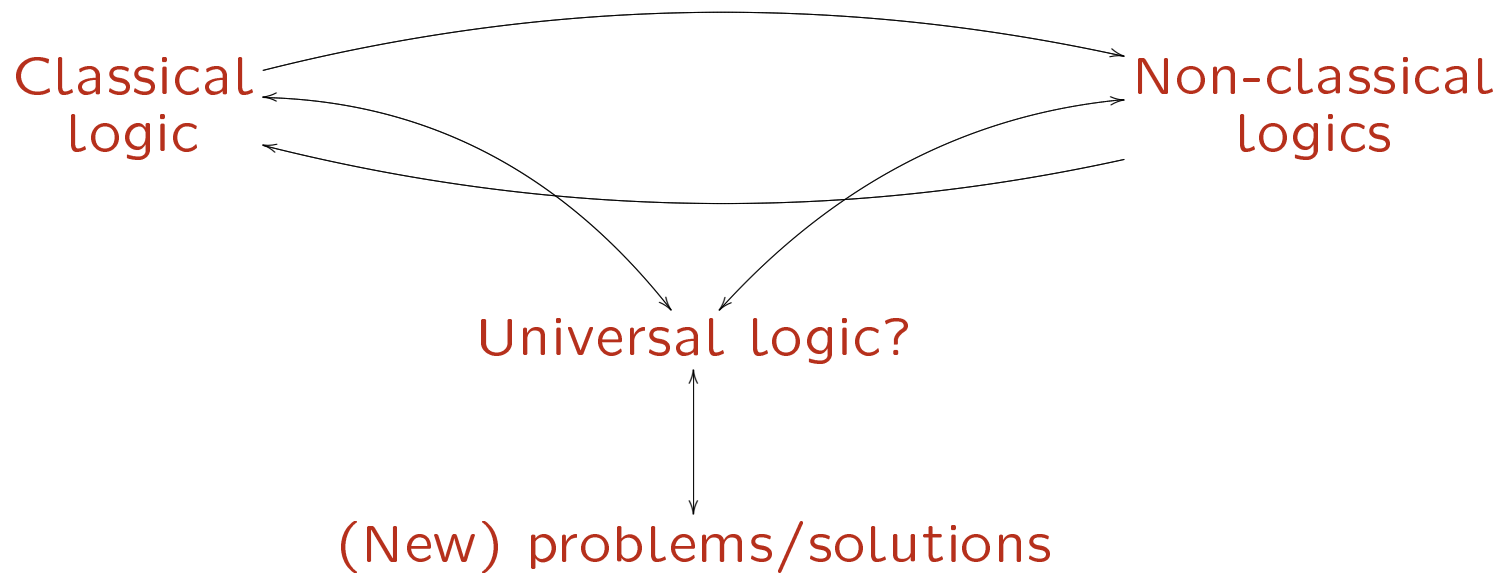
EAFIT University

August 25, 2009

(Last modification: January 7, 2011)

---

# Motivation



---

# Non-classical logics

Definition: ?

Sources: (Non exhaustive!)

- Reject of the logic principles
- Reduction/Expansion of the logical connectives
- Reject of the properties of the consequence relation
- Modifications to the mathematical structure of the consequence relation

---

## Non-classical Logics (cont.)

### Notation:

|                                     |                              |
|-------------------------------------|------------------------------|
| $For :$                             | Set of well-formed formulas  |
| $\alpha, \beta, \delta, \dots :$    | Formulas                     |
| $\Delta, \Gamma, \dots :$           | Theories ( $\subseteq For$ ) |
| $\neg, \wedge, \vee, \rightarrow :$ | logical connectives          |
| $\vdash, \models, \Vdash :$         | Consequence relations        |

---

# Reject of the principle of bivalence

Principle: Every proposition is either true or false

Many-valued logics:<sup>1,2</sup> The number of truth values is not restricted to only two

- Truth values
  - designed
  - anti-designed
  - designed and anti-designed
  - neither designed nor anti-designed
  - no designed
  - no anti-designed

---

<sup>1</sup>N. Rescher. *Many-valued logic*. New York: McGraw-Hill, 1969

<sup>2</sup>L. Peña. *Introducción a las lógicas no clásicas*. Instituto de Investigaciones Filosóficas. Colección: Cuadernos. México D.F.: U. Autónoma de México, 1993

---

## Reject of the principle of bivalence (cont.)

- Semantical universe
  - 0: Minimal element, anti-designed and no designed
  - 1: Maximal element, designed and no anti-designed
  - $\leq$ : partial order
  - $\forall \alpha (0 \leq / \alpha / \leq 1)$ , where  $/ \alpha /$  is the truth-value of  $\alpha$

# Reject of the principle of bivalence (cont.)

Example (Kleene's  $K_3$  logic<sup>3</sup>).

Semantic universe  $\left\{ \begin{array}{l} 1: \text{True (designed)} \\ \frac{1}{2}: \text{Undefined (anti-designed)} \\ 0: \text{False (anti-designed)} \end{array} \right.$

|               | $\neg$        | $\wedge$      | 1             | $\frac{1}{2}$ | 0 | $\vee$        | 1 | $\frac{1}{2}$ | 0             | $\rightarrow$ | 1 | $\frac{1}{2}$ | 0             |
|---------------|---------------|---------------|---------------|---------------|---|---------------|---|---------------|---------------|---------------|---|---------------|---------------|
| 1             | 0             | 1             | 1             | $\frac{1}{2}$ | 0 | 1             | 1 | 1             | 1             | 1             | 1 | $\frac{1}{2}$ | 0             |
| $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ |
| 0             | 1             | 0             | 0             | 0             | 0 | 0             | 1 | $\frac{1}{2}$ | 0             | 0             | 1 | 1             | 1             |

A feature: There is not  $\alpha$  such that  $\models_{K_3} \alpha$

<sup>3</sup>R. L. Epstein. *The Semantic Foundations of Logic: Propositional Logics*. Kluwer Academic Publishers, 1990

---

## Reject of the explosion principle

Principle (pseudo-Scotus, *ex contradictione sequitur quod libet*):

$\forall \Gamma \forall \alpha \forall \beta (\Gamma, \alpha, \neg \alpha \vdash_{CL} \beta)$

Paraconsistent logics:<sup>4,5</sup>  $\exists \Gamma \exists \alpha \exists \beta (\Gamma, \alpha, \neg \alpha \not\vdash_P \beta)$

---

<sup>4</sup>M. A. Bobenrieth. *¿Inconsistencias, por qué no?* Santafé de Bogotá: Tercer Mundo Editores, División Gráfica, 1996

<sup>5</sup>W. A. Carnielli and J. Marcos. A taxonomy of C-systems. In W. A. Carnielli, M. E. Coniglio, and Í. M. L. D'Ottaviano, editors, *Paraconsistency. The logical way to the inconsistent*, pages 1–94. New York: Marcel Dekker, 2002)



---

## Reject of the explosion principle (cont.)

Example (da Costa's  $C_1$  logic).

Bivalent semantic for  $C_1$ :<sup>6</sup>

A valuation for  $C_1$  is a function  $v : For(C_1) \rightarrow \{0, 1\}$  such that:

- $v(\alpha * \beta)$  has classical behavior ( $* \in \{\wedge, \vee, \rightarrow\}$ )
- $v(\alpha) = 0 \Rightarrow v(\neg\alpha) = 1, \quad v(\neg\neg\alpha) = 1 \Rightarrow v(\alpha) = 1$

A consequence: The semantic for  $C_1$  is not truth-functionality ( $v(\alpha) = 1 \not\Rightarrow v(\neg\alpha) = 1, v(\alpha) = 1 \not\Rightarrow v(\neg\alpha) = 0$ ).

---

<sup>6</sup>J. Marcos. Semânticas de traduções possíveis. Master's thesis, U. Estadual de Campinas, 1999, p. 47

---

## Reject of the explosion principle (cont.)

Example (da Costa's  $C_1$  logic (cont.)).

A feature: The logic  $C_1$  admits a strong negation  $\sim \alpha \stackrel{\text{def}}{=} \neg \alpha \wedge \alpha^\circ$ , where  $\circ$  is the well-behavior operator. The negation  $\sim$  is a classical negation.

---

# Reject of the principle of the excluded third

Principle:  $\forall \alpha (\vdash_{CL} \alpha \vee \neg \alpha)$

Intuitionistic logics:<sup>7,8</sup> Proofs are constructions (programs)

---

<sup>7</sup>D. van Dalen. *Logic and structure*. Springer, 4th edition, 2004

<sup>8</sup>M.-H. Sørensen and P. Urzyczyn. *Lectures on the Curry-Howard isomorphism*, volume 149 of *Studies in Logic and the Foundations of Mathematics*. Elsevier, 2006

---

## Reject of the principle of the excluded third (cont.)

The Brouwer-Heyting-Kolmogorov (BHK) interpretation:

| A construction of:              | Consists of:                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------|
| $\alpha_1 \wedge \alpha_2$      | A construction of $\alpha_1$ and a construction of $\alpha_2$                                  |
| $\alpha_1 \vee \alpha_2$        | An indicator $i \in \{1, 2\}$ and a construction of $\alpha_i$                                 |
| $\alpha_1 \rightarrow \alpha_2$ | A method (function) which takes any construction of $\alpha_1$ to a construction of $\alpha_2$ |
| $\perp$                         | There is not construction                                                                      |
| $\neg \alpha$                   | A method (function) which takes any construction of $\alpha$ into a nonexistent object         |

---

## Reject of the principle of the excluded third (cont.)

| A construction of:         | Consists of:                                                                         |
|----------------------------|--------------------------------------------------------------------------------------|
| $\exists x \in U. \phi(x)$ | An element $a \in U$ and a construction of $\phi(a)$                                 |
| $\forall x \in U. \phi(x)$ | A method (function) which takes any element $x \in U$ to a construction of $\phi(x)$ |

---

## Reject of the principle of the excluded third (cont.)

A feature: The proofs by contradiction are invalids

Prove  $\alpha$

$$\frac{\frac{\neg\alpha \quad \vdots}{\perp}}{\neg\alpha \rightarrow \frac{\perp}{\alpha} \equiv \neg\neg\alpha} \quad \text{Invalid!}$$

But, disprove  $\alpha$  is valid

$$\frac{\frac{\alpha \quad \vdots}{\perp}}{\alpha \rightarrow \perp \equiv \neg\alpha}$$

---

# Expansion of the logical connectives

- Modal logics<sup>9</sup>  
( $\Box$  : necessity,  $\Diamond$  : possibility)
  - Temporal logics
  - Epistemic logics
  - Deontic logics

---

<sup>9</sup>G. E. Hughes and M. J. Cressivell. *Introducción a la lógica modal*. Colección: Estructura y Función. Madrid: Editorial Tecnos, 1973

---

# Reduction of the logical connectives

Possible reductions:<sup>10</sup>

- Positive logics
- Implicative logics
- . . .

General question: What is a  $X$ ? ( $X \in \{\neg, \wedge, \vee, \rightarrow\}$ )

---

<sup>10</sup>H. Rasiowa. *An algebraic approach to non-classical logics*, volume 78 of *Studies in Logic and the Foundations of Mathematics*. Amsterdam: North-Holland Publishing Company, 1974

---



---

## A general definition of logic?

**Definition.** A logic  $\mathcal{L}$  is a structure  $\mathcal{L} = \langle For, \Vdash \rangle$  where the consequence relation  $\Vdash \subseteq P(For) \times For$  satisfies:<sup>11,12</sup>

- Reflexivity: If  $\alpha \in \Gamma$ , then  $\Gamma \Vdash \alpha$
- Monotony: If  $\Gamma \Vdash \alpha$  and  $\Gamma \subseteq \Delta$ , then  $\Delta \Vdash \alpha$
- Transitivity: If  $\Gamma \Vdash \alpha$  and  $\Delta, \alpha \Vdash \beta$ , then  $\Gamma, \Delta \Vdash \beta$

---

<sup>11</sup>J.-Y. Béziau. What is paraconsistent logic? In D. Batens, C. Mortensen, G. Priest, and J.-P. van Bendegem, editors, *Frontiers of paraconsistent logic*, pages 95–111. Hertfordshire, England: Research Studies Press LTD., 2000

<sup>12</sup>W. A. Carnielli and J. Marcos. A taxonomy of C-systems. In W. A. Carnielli, M. E. Coniglio, and Í. M. L. D'Ottaviano, editors, *Paraconsistency. The logical way to the inconsistent*, pages 1–94. New York: Marcel Dekker, 2002

---

## Reject of the properties of the consequence relation

- Non-reflexivity logics. Alfabar Logics<sup>13</sup>

**Example.** Let  $\mathcal{L} = \langle \text{For}, \Vdash \rangle$  be a logic such that  $\Gamma \Vdash \alpha$  iff exists  $\Gamma'$  such that  $\Gamma' \subseteq \Gamma$  and  $\Gamma'$  is consistent, and  $\Gamma' \vdash_{CL} \alpha$ . Therefore,  $p \wedge \neg p \not\Vdash p \wedge \neg p$ .

- Non-monotonic logics:

“family of formal frameworks ...in which reasoners draw conclusions tentatively, reserving the right to **retract** them in **the light of further information**.”<sup>14</sup>

- Non-transitive logics: ?

---

<sup>13</sup>D. Krause and J.-Y. Béziau. Relativizations of the principle of identity. *Logic Journal of the IGPL*, 5(3):327–338, 1997

<sup>14</sup>A. Antonelli. Non-monotonic logic. Stanford Encyclopedia of Philosophy, 2006. Eprint: [plato.stanford.edu/entries/logic-nonmonotonic/](http://plato.stanford.edu/entries/logic-nonmonotonic/)

---

# Modifications to the mathematical structure of the consequence relation

Consequence multiples:  $\Vdash \subseteq P(For) \times P(For)$

Sub-structural logics:<sup>15</sup>

- Multi-set  $\neq$  set:  $(\{A, A, B\} \neq \{A, B\})$ , therefore  $\alpha, \alpha, \beta \Vdash \gamma$  does not imply  $\alpha, \beta \Vdash \gamma$
- $\alpha, \beta \Vdash \gamma$  does not imply  $\beta, \alpha \Vdash \gamma$
- In general, a theory  $\Gamma$  has not to be a set

---

<sup>15</sup>G. Restall. *An Introduction to substructural logics*. Taylor & Francis, 2004

---

# Towards an universal logic?

Béziau's approach: What is . . . ?<sup>16,17,18</sup>

Some questions:

- Others approach to the consequence relations (e.g. visual inference)
- Equivalence criteria between semantics, syntax, and algebra for a logic

---

<sup>16</sup>J.-Y. Béziau, R. P. de Freitas, and J. P. Viana. What is classical propositional logic? (a study in universal logic). *Logical Studies*, 7, 2001

<sup>17</sup>J.-Y. Béziau. What is paraconsistent logic? In D. Batens, C. Mortensen, G. Priest, and J.-P. van Bendegem, editors, *Frontiers of paraconsistent logic*, pages 95–111. Hertfordshire, England: Research Studies Press LTD., 2000

<sup>18</sup>J.-Y. Béziau. Are paraconsistents negations negations? In W. A. Carnielli, M. E. Coniglio, and Í. M. L. D'Ottaviano, editors, *Paraconsistency. The logical way to the inconsistent*, pages 465–486. New York: Marcel Dekker, 2002

---

---

## Towards an universal logic? (cont.)

Some questions (cont.):

- Equivalence criteria between logics (e.g. Possible-translation semantics)
- Minimal properties of the logic connectives. (e.g. What is a negation?)
- Compatibility between the logic connectives
- High-order logic extensions

---

## Possible applications

- Mathematical theories construction<sup>19</sup>
- Hypercomputation<sup>20,21</sup>
  - “Or maybe paraconsistent logic will save us from the tricephalous CGC-monster (CGC for Cantor-Gödel-Church) by providing foundations for **finite decidable complete mathematics**.”<sup>22</sup>
- *Type : Type*<sup>23</sup>

---

<sup>19</sup>C. Mortensen. *Inconsistent mathematics*. Kluwer Academic Publishers, 1995

<sup>20</sup>R. Sylvan and J. Copeland. Computability is logic-relative. *Philosophy Research Papers, University of Canterbury*, 5:1–16, 1998

<sup>21</sup>J. C. Agudelo and A. Sicard. Máquinas de Turing paraconsistentes: una posible definición. *Matemáticas: Enseñanza Universitaria*, XII(2):37–51, 2004

<sup>22</sup>J.-Y. Béziau. The future of paraconsistent logic. *Logical Studies*, 2, 1999, p. 16

<sup>23</sup>P. Martin-Löf. A theory of types. Technical report, University of Stockholm, 1971

---

---

## Conclusions

- Tolerance principle in Mathematics (Newton da Costa, 1958):  
“Desde el punto de vista sintáctico-semántico, toda teoría es admisible, desde que no sea trivial. En sentido amplio, *existe, en matemática, lo que no sea trivial.*”<sup>24</sup>
- Logical pluralism<sup>25</sup>
- A new crisis? New opportunities?

---

<sup>24</sup>M. A. Bobenrieth. *¿Inconsistencias, por qué no?* Santafé de Bogotá: Tercer Mundo Editores, División Gráfica, 1996, p. 180

<sup>25</sup>O. Bueno. Can a paraconsistent theorist be a logical monist? In W. A. Carnielli, M. E. Coniglio, and Í. M. L. D'Ottaviano, editors, *Paraconsistency. The logical way to the inconsistent*, pages 535–552. New York: Marcel Dekker, 2002

---

## The category of the logics (bonus slides)

A category  $C$  is given the following data

- A class of objects  $Obj(C)$
- A class of arrows or morphisms  $Mor(C)$
- The functions  $dom, cod : Mor(C) \rightarrow Obj(C)$   
Notation:  $f : A \rightarrow B \equiv f \in Mor(C), dom\ f = A, cod\ f = B$
- For  $A \in Obj(C)$ , the identity arrow  $1_A : A \rightarrow A$
- A composition operator  $\circ : Mor(C) \times Mor(C) \rightarrow Mor(C)$



---

## The category of the Logics (cont.)

These data are subject to the conditions

- $g \circ f$  is defined iff  $\text{cod } g = \text{dom } f$
- If  $g \circ f$  is defined, then  
 $\text{dom}(g \circ f) = \text{dom } f$  and  $\text{cod}(g \circ f) = \text{cod } g$
- For any  $f : A \rightarrow B$ ,  
 $1_B \circ f = f$  and  $f \circ 1_A = f$
- For any  $f : A \rightarrow B, g : B \rightarrow C, h : C \rightarrow D$ ,  
 $h \circ (g \circ f) = (h \circ g) \circ f$

---

## The category of the logics (cont.)

Example (The category **Set**).

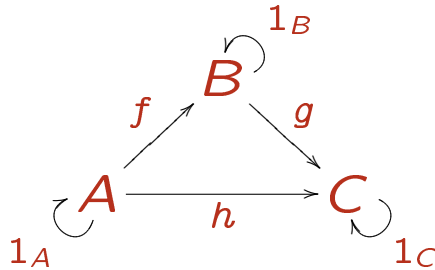
- $Obj(\mathbf{Set})$ : Sets
- $Mor(\mathbf{Set})$ : functions
- The identity arrow  $1_A$ : The identity function
- The composition operator  $\circ$ : The composition of functions

**Technical remark:** The usual definition of a function  $f : A \rightarrow B$  as a set  $f \subseteq A \times B$  which is **single-valued** and **totally defined** is not sufficient to uniquely determine  $cod\ f$ . Therefore it is necessary to define  $f$  as a triple  $(A, graph(f), B)$ .

---

## The category of the Logics (cont.)

Example (The category **3**).



---

## The category of the Logics (cont.)

**Example.** Almost every known example of a mathematical structure with the appropriate structure-preserving map yields a category.

| Category     | Objects            | Morphisms                          |
|--------------|--------------------|------------------------------------|
| <i>Set</i>   | Sets               | Functions                          |
| <i>Pfn</i>   | Sets               | Partial functions                  |
| <i>Vect</i>  | Vector spaces      | Linear transforms                  |
| <i>Top</i>   | Topological spaces | Continuous functions               |
| <i>Poset</i> | Posets             | Monotone functions                 |
| <i>CPO</i>   | Complete posets    | Continuous functions               |
| <i>Lat</i>   | Lattices           | Structure preserving homomorphisms |

---

## The category of the Logics (cont.)

**Example.** A deductive system  $\Vdash_D$  can be turned on a category  $\mathbf{D}$

- $Obj(\mathbf{D})$ : Formulas
- $Mor(\mathbf{D})$ : Proofs
- The identity arrow  $1_A : A \rightarrow A$ : A proof of  $A \Vdash_D A$
- The composition operator  $\circ$ : Transitivity of the  $\Vdash_D$

$$\frac{f : A \rightarrow B \quad g : B \rightarrow C}{g \circ f : A \rightarrow C}$$